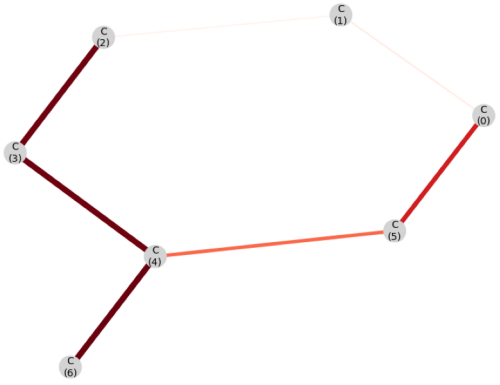
Week Report: MM-GNNExplainer Analysis (MUTAG)

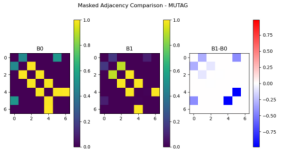
Objective: Analyze the impact of removing Attention (B1), Interaction (B2), and Domain Rules (B3).

1. Qualitative: B0 (Baseline) focuses on C-C bonds in the ring structure (see Semantic Viz).
   B2 (No Interaction) produces a denser mask, highlighting almost all edges (less selective).
2. Quantitative: B2 has significantly higher mask density ($p<0.05$). B0 is more sparse/faithful.
3. Semantic: Top edges for B0 are C-C (Weight > 0.98). B1 drops some C-C edges.

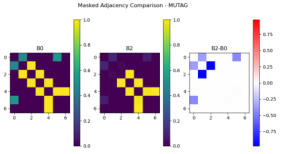
Semantic Interpretation (B0)



mutag B1 vs B0 comparison.png



mutag B2 vs B0 comparison.png



mutag B3 vs B0 comparison.png

